Computer architecture defines how a computer’s hardware components are organized, integrated, and instructed to perform tasks. Types are..

By Instruction Set Architecture (ISA)

RISC (Reduced Instruction Set Computer): Uses a simple, small set of instructions designed to execute in a single clock cycle. It relies heavily on software and registers to achieve high execution speed (e.g., ARM, RISC-V).

CISC (Complex Instruction Set Computer): Uses a large, specialized set of instructions where single complex instructions can execute multi-step operations directly. It reduces the number of instructions per program at the cost of hardware complexity (e.g., x86 architecture).

By Memory Organization

Von Neumann Architecture: Features a single shared memory space and system bus for both instructions and data. Because the CPU cannot read an instruction and transfer data simultaneously, it experiences a throughput limit known as the Von Neumann bottleneck.

Harvard Architecture: Uses physically separate memory units and dedicated buses for instructions and data. This allows simultaneous fetching of instructions and data, significantly improving speed (commonly used in DSPs and embedded systems).

Modified Harvard Architecture: Combines separate instruction and data caches at the CPU level with a unified main memory system, balancing high performance with operational efficiency (standard in modern general-purpose processors).

By Parallelism & Data Processing Stream (Flynn's Taxonomy)

SISD (Single Instruction, Single Data): Executes one instruction stream on one data stream at a time (traditional single-core CPUs).

SIMD (Single Instruction, Multiple Data): Applies a single instruction simultaneously across multiple data points in parallel (used extensively in GPUs, vector processors, and AI hardware).

MISD (Multiple Instruction, Single Data): Processes the same single data stream through multiple instruction pipelines simultaneously (rarely used, mainly in specialized fault-tolerant systems).

MIMD (Multiple Instruction, Multiple Data): Executes multiple independent instructions on distinct data streams simultaneously (found in modern multi-core CPUs, distributed clusters, and supercomputers).